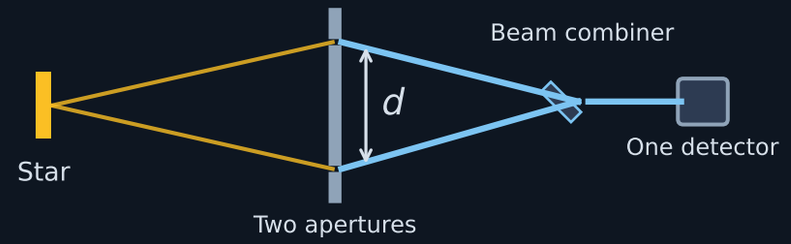
Two ways to measure the angular size of a star

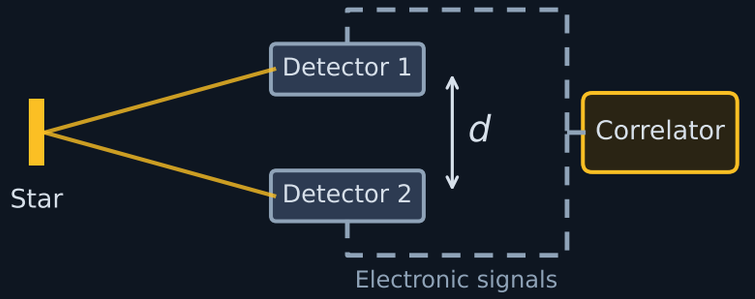
A Michelson — amplitude interferometry

Optical fields are combined

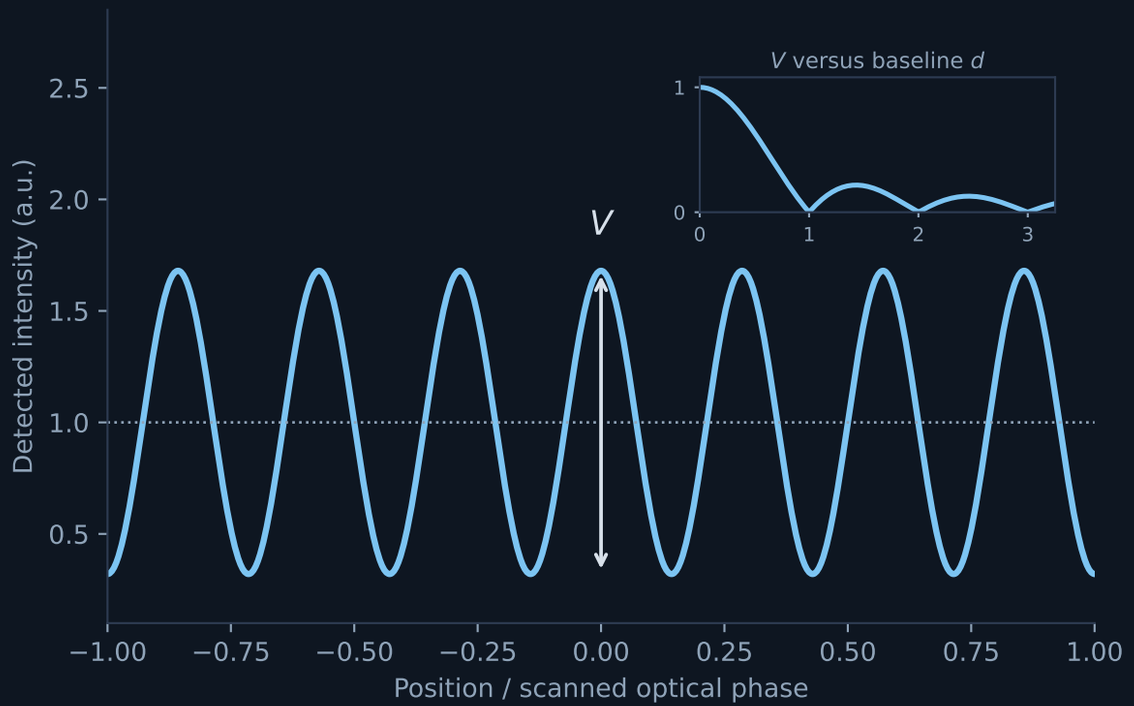


B HBT — intensity interferometry

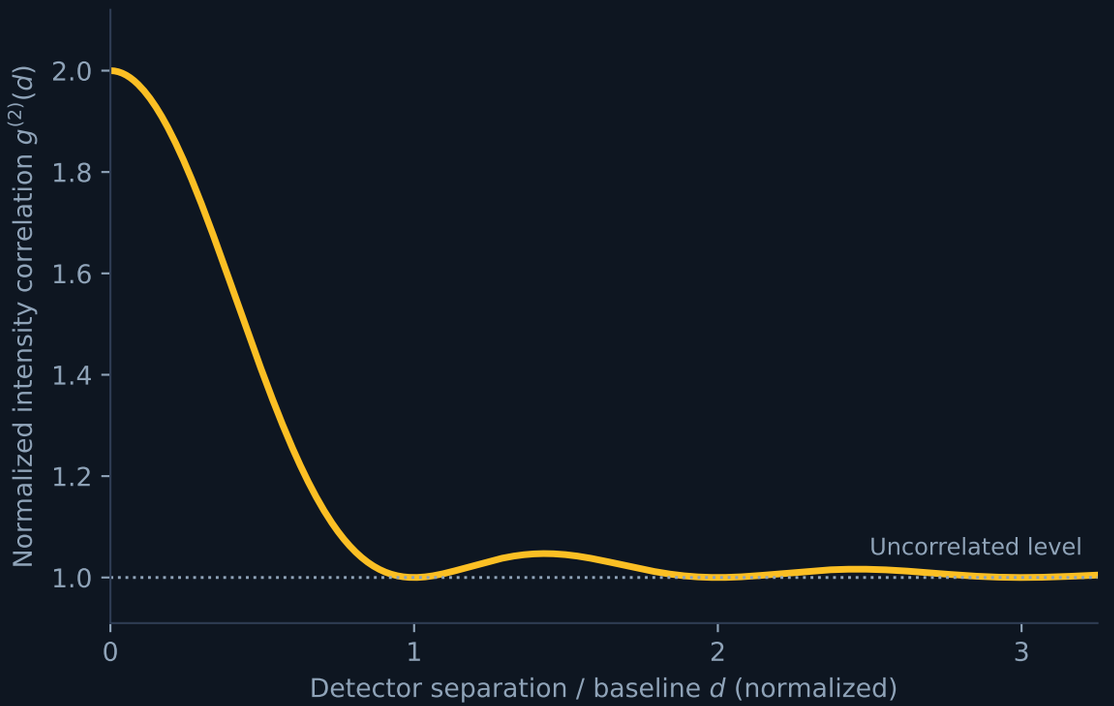
Detector signals are correlated



Fringes: measure their visibility



Correlate intensities at two detectors



Illustrative 1D top-hat star: $V(d) = |\mathrm{sinc}(d)|$ and $g^2(d) = 1 + \mathrm{sinc}^2(d)$ (chaotic light, equal optical paths).